

APPENDIX A

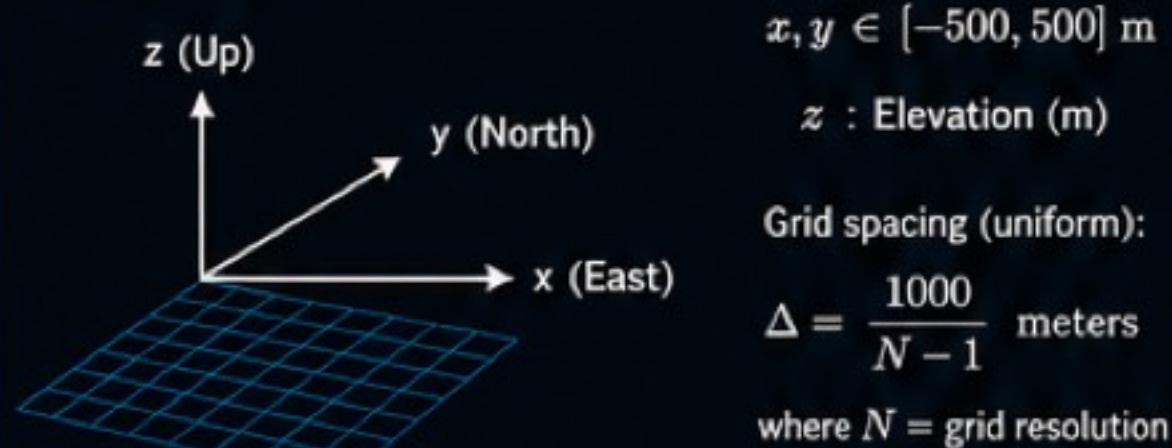
AEROSPACE MATHEMATICS

MODELS, EQUATIONS, AND ALGORITHMS

This appendix documents the mathematical foundations used in the Planetary Autonomous Landing Zone Assessment application, including terrain modeling, environmental scoring, suitability computation, and hazard-aware rover path planning.

1 COORDINATE SYSTEM

A local Cartesian frame is used to model the terrain.
Origin at map center.



3 CRATER & BOULDER MODELS

Crater (negative Gaussian bowl with rim):

$$z_{\text{crater}}(r) = -d e^{-\left(\frac{r}{R}\right)^2} + 0.33 d e^{-\left(\frac{r-R}{0.32R}\right)^2}$$

where:
 $r = \sqrt{(x-x_c)^2 + (y-y_c)^2}$ (radial distance)
 R : Crater radius (m)
 d : Crater depth (m)
 (x_c, y_c) : Crater center (m)

Boulder (positive Gaussian hill):

$$z_{\text{boulder}}(r) = h e^{-\left(\frac{r}{R_b}\right)^2}$$

where:
 R_b : Boulder radius (m)
 h : Boulder height (m)

5 HAZARD PENALTY FIELDS

Hazards (craters and boulders) influence a penalty field that increases with severity.

$$P(x, y) = \sum_k w_k e^{-\left(\frac{d_k(x, y)}{\sigma_k}\right)^2}$$

where:
 $d_k(x, y)$: Distance to feature k
 w_k : Weight proportional to severity
 σ_k : Influence radius (m)

Hazard score (1 = safe, 0 = hazardous):

$$S_{\text{hazard}} = \text{clamp}\left(1 - (0.58 P_{\text{crater}} + 0.42 P_{\text{boulder}}), 0, 1\right)$$

7 OVERALL LANDING ZONE SUITABILITY SCORE

The final Landing Zone (LZ) score is a weighted combination of all factors.

$$S_{LZ} = 100 \times \frac{w_h S_{\text{hazard}} + 0.28 S_{\text{slope}} + w_s S_{\text{solar}} + w_c + 0.10 S_{\text{thermal}}}{w_h + 0.28 + w_s + w_c + 0.10}$$

w_h : Hazard avoidance weight
 w_s : Solar exposure weight
 w_c : Communication weight
(Thermal weight fixed at 0.10)

RISK CLASSIFICATION

● **LOW RISK** $S_{LZ} \geq 85$
● **MODERATE RISK** $70 \leq S_{LZ} < 85$
● **HIGH RISK** $S_{LZ} < 70$

9 ADDITIONAL CONVERSIONS & CONSTANTS

Unit Conversions

1 km = 1000 m
1 m = 1 meter
1 hr = 3600 s
1 deg = $\frac{\pi}{180}$ rad
1 min = 60 s

Body Constants (Mean Values)

Body	Gravity (m/s ²)	Solar Flux (W/m ²)	Radius (km)	Day Length
Moon	1.62	1361	1737.4	29.53 Earth days
Mars	3.71	586	3389.5	24.62 hr
Phobos	0.0057	586	11.27	7.66 hr
Europa	1.315	50	1560.8	3.55 Earth days
Custom	User-defined across parameters			

11 KEY ASSUMPTIONS & LIMITATIONS

Assumptions

- Users provide accurate mission inputs.
- Procedural terrain approximates real regolith terrain.
- Illumination model assumes clear-sky conditions.
- Communication model is geometry-based.
- Rover operates on nominal terrain conditions.

Limitations

- Not a flight-certified system.
- Does not model lander plume interaction.
- Does not model regolith mechanics or bearing strength.
- Does not model extreme thermal or radiation environments.
- Not a substitute for high-fidelity mission analysis.

2 TERRAIN MODEL GENERATION

The synthetic terrain elevation is modeled as:

$$z(x, y) = \sum_i A_i \sin\left(\frac{x}{\lambda_{x,i}}\right) + \sum_j B_j \cos\left(\frac{y}{\lambda_{y,j}}\right) + \sum_k C_k \sin\left(\frac{x+y}{\lambda_{d,k}}\right) + e(x, y) + z_{\text{body}}(x, y) + z_{\text{craters}} + z_{\text{boulders}}$$

A_i, B_j, C_k : Amplitudes (m)

$\lambda_{x,i}, \lambda_{y,j}, \lambda_{d,k}$: Spatial wavelengths (m)

$e(x, y)$: Zero-mean Gaussian noise $\sim \mathcal{N}(0, \sigma^2)$

$z_{\text{body}}(x, y)$: Body-specific terrain perturbation (see right)

$z_{\text{craters}}, z_{\text{boulders}}$: Localized features (see Section 3)

σ : Noise standard deviation

BODY PERTURBATIONS $z_{\text{body}}(x, y)$

🌕 Moon	= 0
🔴 Mars	= $5 \sin\left(\frac{x}{75}\right)$
🔴 Phobos	= $26 \sin\left(\frac{x}{260}\right) + 19 \cos\left(\frac{y}{240}\right)$
🌐 Europa	= $7 \sin\left(\frac{x - 0.5y}{55}\right)$
⚙️ Custom	User-defined

4 SLOPE CALCULATION

Terrain slope at each grid cell is computed from the gradient of the elevation field.

$$\frac{\partial z}{\partial x} \approx \frac{z_{i,j+1} - z_{i,j-1}}{2\Delta}$$

$$\frac{\partial z}{\partial y} \approx \frac{z_{i+1,j} - z_{i-1,j}}{2\Delta}$$

$$\theta = \tan^{-1} \left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \right)$$

θ : Slope angle (degrees)

Δ : Grid spacing (m)

SLOPE SCORE

$$S_{\text{slope}} = \text{clamp}\left(1 - \frac{\theta - \theta_{\text{soft}}}{\theta_{\text{hard}} - \theta_{\text{soft}}}, 0, 1\right)$$

$\theta_{\text{soft}}, \theta_{\text{hard}}$: Soft and hard slope limits (deg)

6 ENVIRONMENTAL SCORE MODELS

Solar (Illumination) Score

$$S_{\text{solar}} = I \times \left(\frac{F_{\text{solar}}}{1361}\right)$$

• $I = \max(\vec{n} \cdot \vec{s}, 0)$: Surface incidence (0-1)

• F_{solar} : Body solar flux (W/m²)

• $\vec{n}$: Surface normal, $\vec{s}$: Sun unit vector

Terrain shadowing reduces S_{solar} by factor 0.08 when in shadow.

Communication (Line-of-Sight) Score

$$S_{\text{comm}} = (0.55 B + 0.45 E) P_{\text{body}}$$

• B : Elevation geometry / LOS term (0-1)

• E : Normalized elevation gain (0-1)

• P_{body} : Body link penalty factor (see table)

Blocked LOS reduces S_{comm} by factor 0.15.

Thermal Score

$$S_{\text{thermal}} = 1 - P_{\text{thermal}} \quad P_{\text{thermal}} : \text{Body thermal penalty (see table on right)}$$

8 ROVER PATH PLANNING (HAZARD-AWARE)

Cost Map for Path Planning

$$C = 1 + \frac{100 - S_{LZ}}{22} + C_{\text{slope}} + C_{\text{shadow}}$$

• C_{slope} : Slope penalty (0-3)

• C_{shadow} : Shadow penalty (0 or 0.45)

• C_{comm} : LOS blocked penalty (0 or 0.55)

Higher cost = higher risk / effort.

A* Search Algorithm

$$f(n) = g(n) + h(n)$$

• $g(n)$: Cost from start to node n

• $h(n)$: Heuristic (Euclidean distance)

• 8-connected grid (N, S, E, W, NE, NW, SE, SW)

• Returns minimum-cost path.

Waypoint Pattern (Survey)

Star pattern about the LZ



Variable radius r

Path Metrics

• Path length $L = \sum_{i=0}^{M-2} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}$ P_{thermal} : Body thermal penalty (see table on right)

10 AI ADVISORY LAYER (NON-AUTHORITATIVE)

Assessment Pipeline

Physics Engine (Authoritative)

Terrain, scores, paths, metrics

LLM Advisor (Advisory)

Hazard brief, recommendations

Mission Report (Combined)

Physics results + advisory text (LLM does not modify results)

LLM provider, model, and prompt version are user-configurable (if enabled).

Note

This application uses simplified models and surrogate metrics for conceptual mission analysis and decision support. Use mission-specific data, validated tools, and detailed analysis for flight-critical planning.